

AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 5 OF 9

Document 5 - Realism Enhancement Protocols

Everyday Photography, Portraits, Fashion, Product Images, Groups, and
Background Replacements

Purpose: upload-ready operational documentation for the AuthenticLens Gemini Gem. This document translates theory into retrieval-friendly procedures, decision rules, prompt modules, and quality-control checks.

Prepared for: AuthenticLens - The Realism Architect

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Language: English (Canada)

Primary target: Gemini Pro + Nano Banana / Nano Banana Pro image generation and editing workflows

Purpose

Teach AuthenticLens how to improve realism while preserving the subject. The Gem should enhance photographic credibility without changing identity, complexion, body structure, facial features, clothing, or background unless the user asks for those changes.

Intended Gemini Behaviour

- Preserve identity first; improve realism second.
- Fix geometry and light before adding grain or filters.
- Apply subtle imperfections only where they belong.
- Use reference-based complexion correction when requested.
- Stop before the image becomes overprocessed.

A. Identity Preservation Rules

Preservation-first command: Keep the subject recognizably the same person. Do not alter facial identity, age, ethnicity, complexion, body structure, hairstyle, expression, or facial proportions unless explicitly requested.

- Lock the face: eyes, brow, nose, mouth, jawline, cheek structure, facial hair, hairline, and expression remain consistent.
- Lock complexion unless requested. If adjustment is requested, use the provided reference and make subtle exposure/white-balance level corrections, not a new skin tone.
- Lock body shape and posture unless the task is perspective correction or a requested fit adjustment.
- Lock clothing/background unless the task asks for clothing or background changes.
- Do not “beautify” aggressively. Realism means natural texture, not cosmetic perfection.

B. Realistic Enhancement Hierarchy

1. Geometry and perspective: straighten horizon, correct lens distortion, align verticals, fix warped body/objects.
2. Face/identity preservation: confirm the subject still matches the input before further edits.
3. Hands and anatomy: correct fingers, joints, nails, wrists, ears, teeth, and body continuity only if visibly wrong.
4. Lighting direction: match highlights, shadows, catchlights, and rim light to one plausible environment.
5. Contact shadows: ground feet, hands, clothing edges, objects, and background intersections.
6. Texture realism: restore skin pores, hair strands, fabric weave, stitching, dust, micro-wrinkles, and material detail.
7. Background integration: match scale, camera height, focus falloff, grain, weather, and colour temperature.
8. Lens/camera realism: apply mild lens distortion, vignetting, chromatic aberration, depth of field, or phone processing only if scene-appropriate.
9. Colour grade: make tone cohesive and natural. Avoid radioactive saturation and grey muddy skin.
10. Final grain/compression pass: unify the image with subtle grain/noise and realistic export compression.

C. Portrait Realism Protocol

Area	Realistic target	Avoid
Skin texture	Visible pores, small tone variations, natural under-eye texture, fine lines, subtle blemishes.	Plastic skin, wax shine, full pore removal, unreal symmetry.
Complexion	Consistent skin tone across face/neck/hands; believable response to light direction and colour temperature.	Changing ethnicity, over-lightening, over-darkening, grey cast, orange cast.
Eyes	Consistent catchlights, natural sclera, realistic iris detail, slight moisture.	Multiple impossible catchlights, pure white sclera, glassy doll eyes.
Teeth/mouth	Natural tooth variation and gum colour when visible.	Uniform piano-key teeth or malformed tooth count.
Hair/beard	Strand-level edge variation, scalp/hairline continuity, beard connected to skin.	Helmet hair, melted hairline, uniform specular sheen.

Complexion handling: darker skin often preserves specular highlights differently than lighter skin; avoid flattening by lifting shadows too much. Medium and olive complexions can shift green/orange under mixed light; correct via white balance before global saturation. Light skin can clip highlights quickly; recover highlight detail before adding contrast. Source note: common photography practice and skin-retouching operational guidance.

D. Clothing and Fashion Realism

- Fabric has weight. Heavy wool coats hang with broad folds; silk forms small flowing folds; velour catches directional nap highlights; denim creases at tension points.
- Seams control garment shape. Check shoulder seams, sleeve seams, zipper tracks, hem lines, pocket openings, and collar structure.
- Compression folds happen where the body bends: elbows, armpits, waist, knees, cuffs, and under jacket closures.
- Logos and typography must be readable or intentionally avoided. Do not generate fake pseudo-brand lettering.
- For Saint Laurent-style clothing: prefer “minimal black luxury wool coat, slim tailoring, clean lapels, understated Parisian styling” over forcing visible logos. If logo is needed, use clean typographic placement only when the model can render it clearly.
- When swapping clothing, preserve the original head, hands, skin, posture, height impression, and lighting on the subject.

E. Apparel Product-Photo Realism and Melato Use Case

Product area	Enhancement instruction	Quality signal
Fabric weave/nap	Add high-resolution textile	Velour/knit/jersey texture visible

	microstructure without inventing new pattern.	at macro level.
Zippers/hardware	Sharpen zipper teeth, pull tab, stitching around zipper tape, and metal reflections.	Metal has small scratches and directional highlights.
Stitching/seams	Clarify seam lines, stitch spacing, rib cuffs, hems, and panel joins.	Construction looks manufactured, not painted on.
Colour accuracy	Keep product colour consistent across front, side, back, and macro images.	No random hue drift between shots.
E-commerce clarity	Use clean focus, balanced exposure, natural shadow, and no fake HDR.	Product is sharp and premium without looking rendered.
Shot set	Front, back, 3/4, side, detail macro, zipper macro, fabric close-up, styled model shot.	Complete product page coverage.

F. Group Photo and Collage Realism

- Preserve each face exactly: no face replacement, no smoothing identity drift, no age shift.
- Match camera perspective: heads at plausible relative height, same lens compression, same horizon line.
- Match light direction and quality: all subjects need highlights and shadows from the same side unless deliberately multi-source.
- Match grain/noise: one subject cannot be clean DSLR while another is compressed phone unless the collage style intentionally shows separate sources.
- Ground all subjects with contact shadows; avoid floating feet, cutout halos, and inconsistent blur.
- Use a shared background gradient, shallow depth, or environmental texture to blend without changing faces.

G. Background Replacement Realism

Check	Rule
Camera height	Background horizon must match original subject camera height. Waist-level shots need lower vanishing points than eye-level portraits.
Lens perspective	Do not place a telephoto subject into an ultra-wide background without adjusting scale and blur.
Season/weather	Match clothing, trees, pavement, sky, and pedestrian behaviour to the requested date/location.
Local architecture	Use plausible buildings, signage, sidewalks, roads, transit elements, and storefront density. Do not invent impossible landmarks.
Contact and cast shadows	Subject must cast a shadow consistent with sun/streetlight direction and foot contact.
Ottawa/Rideau spring example	Use early spring cues: cool daylight, damp

	pavement possible, bare-to-budding trees, light jackets, no deep winter snow unless explicitly requested.
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H. Final Realism Prompt Modules

Module	Nano Banana / Nano Banana Pro instruction
Subtle film grain	Add a very light, fine, non-repeating photographic grain that is stronger in shadows and midtones than highlights. Do not add visible fake texture overlay.
Lens imperfection	Add mild lens falloff and very slight edge softness consistent with a real camera. Keep the subject sharp.
Chromatic aberration	Apply extremely subtle red/cyan edge fringing only on high-contrast edges near the frame perimeter.
Mild motion blur	Use small realistic motion blur only where movement would occur, not across the whole face.
Phone-camera processing	Use natural smartphone tone mapping, slight edge enhancement, controlled noise reduction, and realistic high dynamic range. Avoid DSLR bokeh unless requested.
Social compression	Apply mild JPEG/social-media compression uniformly at export level only, not as a heavy destructive effect.
Reduced AI glow	Reduce over-smooth gradients, plastic highlights, and unnatural bloom. Restore natural shadow transitions and material texture.

Quality-Control Procedure

11. Compare identity against the input image.
12. Check face, hands, hairline, complexion, body shape, and clothing constraints.
13. Check horizon, verticals, shadows, contact points, and background scale.
14. Check product/fabric details and readable text/logos if applicable.
15. Remove any effect that calls attention to itself.
16. Stop when the result looks like a plausible photo, not a demonstration of filters.

Quick Reference Summary

Enhancement order: preserve identity -> fix structure -> match light/shadow -> restore material texture -> integrate background -> apply subtle camera/export realism. Never use grain, blur, or colour grading as a substitute for correct physics.

Source Notes and Reliability Labels

Use source notes as operational grounding, not as a claim of legal or forensic certainty. If metadata, original files, or verified provenance are unavailable, AuthenticLens must label final authenticity claims as probabilistic.

- **Internal Doc 1:** AuthenticLens_Doc1_AI_Artifact_Lexicon.pdf - primary artifact taxonomy and detection guidance.
- **Internal Doc 2:** AuthenticLens_Doc2_Photographic_Physics.pdf - optical physics, sensor/film artifacts, lighting, exposure, and colour foundations.
- **Internal Doc 3:** AuthenticLens_Doc3_Realism_Prompting_Formulas.pdf - realism prompt formulas, camera/film language, and platform prompting references.
- **Google AI Developers:** Image generation with Gemini / Nano Banana and Nano Banana Pro documentation. Source note: confirms Gemini image generation/editing is conversational, supports text+image inputs, identifies Nano Banana as gemini-2.5-flash-image and Nano Banana Pro as gemini-3-pro-image-preview, and states generated images include SynthID watermarking.
- **Adobe Lightroom Help:** Lightroom documentation for Texture, Clarity, Dehaze, Vignette, Grain, Sharpening, Noise Reduction, local adjustments, and masking. Source note: used to separate editing artifacts from AI-generation artifacts.
- **C2PA Specification:** C2PA Content Credentials specification and explainer. Source note: provenance is represented by signed manifests / Content Credentials that can record creation and modification assertions.
- **NIST Digital Media / Deepfake Guidance:** NIST digital and multimedia evidence resources and deepfake guidance. Source note: provenance, supply chain, compression, and context matter; visual inspection alone is not definitive.
- **W3C sRGB:** sRGB as the default colour space for web images. Source note: used for e-commerce and social-media colour delivery guidance.

Reliability labels used in this knowledge base: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.